

R Practice 3 – Functions

This assignment is just for your own practice and does not need to be submitted. If you have any questions, please feel free to reach out to me, come to office hours, or schedule another a time to meet with me.

Open an R Script in RStudio and complete each of the following:

1. **Temperature conversion.**

- a. Write a function named **celsius_to_fahrenheit** that takes a numeric input representing a temperature in Celsius and returns the equivalent temperature in Fahrenheit using the formula: $F = \frac{9}{5} \cdot C + 32$.

- b. Test the function by converting the temperature 25°C to Fahrenheit.

2. **BMI.** Body mass index (BMI) is computed with the formula $BMI = \frac{\text{Weight in kilograms}}{(\text{Height in meters})^2}$. Write a function called **calculate_bmi** with the following properties:

- The function should have two inputs: **height** and **weight**.
- The function should compute the person's BMI and return the BMI value.

Test the function for a person with a height of 1.71 meters and a weight of 59.8 kgs.

3. **Reading documentation.** Type **?plot** to bring up the documentation for the **plot()** function. Select the link for the one in the graphics library. Read a little bit about this function. Try to mess around a bit to make a couple scatterplots.